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Explainable Deep Learning System for Chest X-Ray Classification Using Grad-CAM

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Abstract

Pneumonia is one of the leading causes of mortality worldwide, particularly among children under five and immunocompromised adults. Timely and accurate diagnosis from chest X-ray images is critical, yet remains limited by radiologist availability, diagnostic variability, and access disparities, particularly in low-resource settings. This project presents an explainable deep learning framework for automated pneumonia detection from chest X-ray images, designed to assist clinical decision-making rather than replace physician judgment.

A pretrained ResNet-50 convolutional neural network is fine-tuned on the Kaggle Chest X-Ray dataset, comprising 5,863 labeled images from the Guangzhou Women and Children's Medical Center, using a two-phase transfer learning strategy. The model first trains a custom classification head while the backbone remains frozen, then fine-tunes all layers with differential learning rates and early stopping. Data augmentation and weighted sampling are applied to address class imbalance.

The system achieves 90.1% accuracy, a ROC-AUC of 0.9614, and a weighted F1-score of 0.90 on the held-out test set of 624 images. Gradient-weighted Class Activation Mapping (Grad-CAM) is integrated to generate heatmap overlays that visually highlight the lung regions most influential to each prediction, making the model interpretable to clinical users. The final system is deployed as an interactive Gradio web application on Hugging Face Spaces, enabling real-time inference with explainability overlays from any browser.

1. Introduction

1.1 Background

Medical imaging plays a central role in modern diagnostic medicine. Chest X-rays are among the most frequently ordered imaging studies globally, used to screen for pneumonia, tuberculosis, pulmonary edema, cardiomegaly, and other thoracic conditions. Despite their widespread use, accurate interpretation requires specialized training that is not universally available. In high-volume clinical environments, radiologist fatigue and cognitive load introduce diagnostic variability, while in lower-income regions, a severe shortage of trained radiologists leaves large populations without reliable screening.

Artificial intelligence, and specifically deep learning, has demonstrated remarkable potential in automating the analysis of medical images. Convolutional neural networks (CNNs) trained on large-scale radiographic datasets have achieved diagnostic accuracy comparable to or exceeding that of experienced clinicians in several studies. However, clinical adoption has been slow, in part because

most deep learning models function as black boxes, offering a classification output without any indication of the image features that drove the decision. In healthcare, this opacity is not merely a usability concern; it is a patient safety issue.

1.2 Problem Statement

The core challenge this project addresses is twofold. First, automated chest X-ray classification systems must be accurate enough to be clinically useful. Second, they must be interpretable: a clinician must be able to verify that the model is attending to medically relevant findings rather than spurious image artifacts or demographic confounders. Existing literature demonstrates strong classification performance but often treats interpretability as secondary. This project explicitly positions explainability as a first-class design requirement, integrating Grad-CAM visualization into the inference pipeline and evaluating the clinical coherence of the resulting heatmaps.

1.3 Objectives

- Develop a binary chest X-ray classifier (Normal vs. Pneumonia) using transfer learning on ResNet-50
- Implement a two-phase training strategy to preserve pretrained features while adapting the model to the target domain
- Integrate Grad-CAM to generate spatially meaningful heatmaps for every prediction
- Evaluate model performance using accuracy, precision, recall, F1-score, and ROC-AUC
- Perform systematic error analysis using Grad-CAM on misclassified cases
- Deploy the complete system as a publicly accessible Gradio web application on Hugging Face Spaces

1.4 Contributions

The key technical contributions of this project are:

- Designed and implemented an end-to-end explainable pneumonia detection pipeline using ResNet-50 and Grad-CAM, trained and evaluated on a publicly available clinical dataset
- Implemented a two-phase transfer learning strategy with differential learning rates and early stopping, achieving 90.1% classification accuracy and ROC-AUC of 0.9614
- Integrated Grad-CAM as a first-class module within the inference pipeline, enabling per-prediction visual explanations without architectural modification
- Conducted structured error analysis using Grad-CAM on true positives, false positives, and false negatives, revealing interpretable failure modes linked to viral pneumonia and vascular confusion

- Built and deployed a public-facing Gradio web application providing real-time inference with explainability overlays

1.5 Scope

This project focuses on binary classification of chest X-rays into Normal and Pneumonia categories using the Kaggle Chest X-Ray dataset. The scope does not include multi-class pathology detection, image segmentation, or 3D volumetric imaging. The deployment target is a research-grade web demonstration rather than a certified clinical decision support system. All experiments were conducted on the Google Colab platform using a free-tier NVIDIA Tesla T4 GPU.

2. Literature Review

2.1 CNN-Based Chest X-Ray Classification

The application of convolutional neural networks to chest radiograph classification has progressed rapidly since the release of large-scale annotated datasets. Wang et al. [4] introduced ChestX-ray8, a dataset of over 100,000 frontal-view chest X-rays annotated with 14 disease labels, and demonstrated that CNNs could achieve AUC scores exceeding 0.7 across most categories. Rajpurkar et al. [3] proposed CheXNet, a 121-layer DenseNet trained on the ChestX-ray14 dataset, which achieved a pneumonia F1-score of 0.435, surpassing the mean F1-score of 0.387 achieved by a panel of four radiologists. This landmark result established that deep learning could match or exceed human performance on specific radiographic tasks.

He et al. [1] introduced ResNet (Residual Network), which resolved the degradation problem in very deep networks through the use of shortcut skip connections. ResNet-50, the variant used in this project, has become a standard backbone for medical imaging transfer learning due to its favorable balance of depth, parameter efficiency, and representational capacity. EfficientNet variants have also shown strong results [5], though they are more sensitive to learning rate scheduling and batch size, making them less suitable for rapid prototyping under compute constraints.

2.2 Transfer Learning in Medical Imaging

Transfer learning from ImageNet pretraining has been shown to substantially improve performance in medical imaging contexts where labeled data is scarce. Esteva et al. [5] provided a comprehensive review of deep learning in healthcare, noting that ImageNet-pretrained features transfer effectively to medical domains due to shared low-level visual structure. Raghu et al. [6] specifically investigated transfer learning for medical imaging, finding that fine-tuning pretrained models consistently outperformed training from scratch, even when source and target domains appeared dissimilar.

The two-phase fine-tuning strategy employed in this project follows the approach recommended by Howard and Ruder [7] in the context of natural language processing and later validated in computer vision. Differential learning rates across network layers, with smaller rates for backbone layers and larger rates for task-specific heads, allow gentle adaptation without overwriting pretrained representations.

2.3 Explainable AI in Medical Imaging

The need for interpretability in clinical AI systems is widely recognized. Tonekaboni et al. [8] surveyed clinician attitudes toward AI decision support, finding that transparency in model reasoning was a primary factor in clinical trust and adoption. Selvaraju et al. [2] introduced Gradient-weighted Class Activation Mapping (Grad-CAM), a technique that computes a weighted combination of feature maps

in the final convolutional layer using gradients backpropagated from the target class score. The resulting heatmap is class-discriminative and spatially localizing without requiring architectural modification or retraining.

Arun et al. [9] evaluated multiple saliency methods on chest X-rays and found Grad-CAM to provide the most clinically coherent localizations compared to alternatives including LIME, SHAP, and integrated gradients. This finding supports the selection of Grad-CAM as the explainability method for the present work.

2.4 Limitations of Prior Work

Several limitations in prior systems are noteworthy. Many studies use the ChestX-ray14 dataset which contains label noise due to automated NLP extraction of radiology reports. Binary classification systems, including this project, do not generalize to the multi-label multi-pathology setting. Grad-CAM localizations are sensitive to the choice of target layer and the presence of batch normalization. Most published systems are evaluated under controlled experimental conditions and have not been validated in prospective clinical deployments.

3. System Architecture

3.1 Overall Workflow

The system is organized as a sequential pipeline from raw dataset images to a deployed web application. The five major stages are: (1) data ingestion and preprocessing, (2) model training with two-phase transfer learning, (3) quantitative evaluation, (4) Grad-CAM explainability generation, and (5) Gradio application deployment. Figure 1 illustrates this end-to-end architecture, showing the data flow path (navy), explainability branch (purple), and deployment path (red).

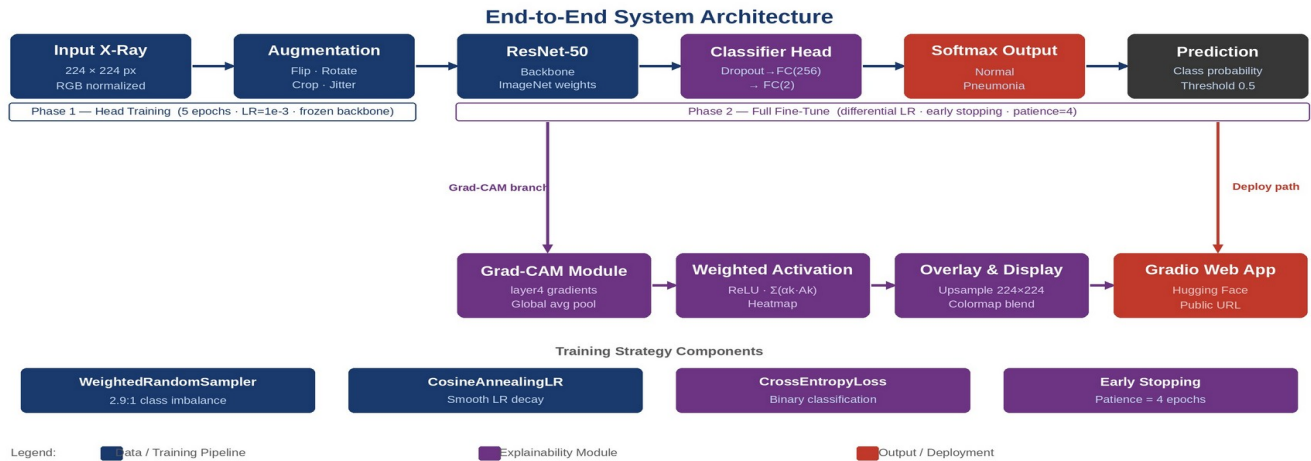


Figure 1: End-to-end system architecture. Navy path shows the main classification pipeline. Purple path shows the Grad-CAM explainability branch. Red path shows deployment output. Bottom row shows training strategy components.

3.2 Data Preprocessing Pipeline

The preprocessing pipeline applies the following transformations, with augmentation applied to training data only:

- Resize to 244 x 244 pixels (training) or 224 x 224 pixels (validation/test)
- Random crop to 224 x 224 (training augmentation)
- Random horizontal flip with probability 0.5 (training only)
- Random rotation within +/- 10 degrees (training only)
- Color jitter: brightness and contrast adjustment within +/- 20% (training only)
- Normalize using ImageNet channel statistics: mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]
- WeightedRandomSampler applied to training DataLoader to compensate for 2.9:1 class imbalance

3.3 Training Pipeline

1. Load pretrained ResNet-50 weights (ImageNet V2 checkpoint)
2. Freeze all backbone parameters; randomly initialize custom classification head
3. Phase 1: Train head only for 5 epochs at LR = $1e-3$
4. Phase 2: Unfreeze all layers; apply differential LRs (backbone: $1e-4$, head: $1e-3$)
5. Save checkpoint to Google Drive after every epoch improvement on validation accuracy
6. Apply early stopping with patience = 4 to prevent overfitting

3.4 Prediction Pipeline

At inference time, a new chest X-ray image is resized and normalized using the validation transforms, passed through the loaded model checkpoint, and a softmax over the two output logits produces class probabilities. The predicted class is the argmax of the probability vector. Grad-CAM is simultaneously computed and overlaid on the input image for display in the Gradio interface.

3.5 Explainability Module

The Grad-CAM module hooks into ResNet-50's layer4, the final convolutional stage before global average pooling, which provides the highest-level spatial feature representation at a 7×7 spatial resolution. For a given input and target class, it computes the gradient of the class score with respect to each spatial location in the feature map, global-average-pools those gradients to obtain importance weights, and produces a weighted sum of feature maps rectified by ReLU. The resulting activation map is upsampled bilinearly to 224×224 and blended with the original image as a color heatmap using the pytorch-grad-cam library.

4. Dataset Description

4.1 Source

The Kaggle Chest X-Ray Images (Pneumonia) dataset was used for this project. Images were originally collected from retrospective cohorts of pediatric patients aged one to five years at the Guangzhou Women and Children's Medical Center, Guangzhou, China. All chest X-ray imaging was performed as part of routine clinical care. The quality of images was graded by two expert physicians before being cleared for use in AI model training, and a third expert reviewed the evaluation set independently.

4.2 Dataset Statistics

Split	Normal	Pneumonia	Total
Train	1,341	3,875	5,216
Validation	8	8	16
Test	234	390	624

Table 1: Dataset split statistics for the Kaggle Chest X-Ray Images (Pneumonia) dataset

4.3 Class Imbalance

The training set exhibits a significant class imbalance: 3,875 pneumonia samples versus 1,341 normal samples, a ratio of approximately 2.9:1. Left unaddressed, a model trained on this distribution would be biased toward predicting pneumonia for all inputs, achieving artificially high accuracy while performing poorly on normal cases. This was addressed through PyTorch's WeightedRandomSampler, which assigns each training sample a weight inversely proportional to its class frequency.

4.4 Ethical Considerations

The dataset was released for research and educational use under the Kaggle dataset license. All images were de-identified prior to public release. The pediatric patient population introduces considerations around data sensitivity; the dataset has been reviewed and used extensively in the published literature. No additional data collection, patient recruitment, or clinical trials were conducted as part of this project. The system is presented as a research prototype and not a certified medical device. Deployment as a clinical tool would require regulatory approval and prospective clinical validation.

5. Methodology

5.1 Model Architecture

The base architecture is ResNet-50, introduced by He et al. [1]. ResNet-50 consists of 48 convolutional layers organized into four stages, each containing bottleneck residual blocks that use 1x1 convolutions to reduce and restore dimensionality. Shortcut connections allow gradients to flow directly from later to earlier layers, resolving the vanishing gradient problem that limited the depth of prior architectures.

The pretrained ResNet-50 classification head was replaced with a custom classifier:

- Dropout($p = 0.4$)
- Linear(2048 to 256)
- ReLU activation
- Dropout($p = 0.3$)
- Linear(256 to 2) [Normal, Pneumonia]

The two dropout layers provide regularization, reducing overfitting on the relatively small training set. The intermediate 256-unit layer adds representational capacity between the backbone features and the binary output.

5.2 Rationale for ResNet-50

ResNet-50 was selected over alternative architectures for three reasons. Its residual connections provide stable gradient flow during fine-tuning, particularly when unfreezing deep layers. It has been validated extensively in medical imaging literature, notably as the direct predecessor to the DenseNet used in CheXNet [3]. Relative to deeper variants such as ResNet-101 and ResNet-152, ResNet-50 provides comparable representational capacity with lower compute requirements, which is important given the Colab free-tier GPU constraint.

5.3 Transfer Learning Strategy

Transfer learning from ImageNet pretraining was employed because the training dataset of 5,216 images is insufficient to train a 50-layer network from scratch without severe overfitting. ImageNet-pretrained low-level features, including edge detectors, texture filters, and shape detectors, are known to transfer effectively to medical imaging tasks. The two-phase strategy proceeds as follows:

- Phase 1 (5 epochs): All backbone parameters are frozen. Only the custom classification head is trained at $LR = 1e-3$. This prevents large gradients from the randomly initialized head from corrupting pretrained weights during early training.

- Phase 2 (up to 10 epochs, with early stopping): All parameters are unfrozen. Backbone layers receive a learning rate of 1e-4 (ten times smaller than the head rate of 1e-3), allowing gentle adaptation without overwriting learned representations.

5.4 Hyperparameters

Hyperparameter	Value
Input Image Size	224 × 224 pixels
Batch Size	32
Phase 1 Learning Rate	1e-3 (classifier head only)
Phase 2 LR — Backbone	1e-4
Phase 2 LR — Head	1e-3
Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.999$)
Loss Function	CrossEntropyLoss
LR Scheduler	CosineAnnealingLR
Phase 1 Epochs	5
Phase 2 Max Epochs	10 (early stopping)
Early Stopping Patience	4 epochs
Dropout Rate	0.4 (layer 1), 0.3 (layer 2)
FC Hidden Units	256
Training Hardware	NVIDIA Tesla T4 GPU (Google Colab)

Table 2: Hyperparameter configuration for model training

5.5 Training Environment

All training was conducted on Google Colaboratory using a free-tier NVIDIA Tesla T4 GPU (16 GB VRAM). Model checkpoints, training curves, evaluation plots, and Grad-CAM outputs were saved to Google Drive for persistence across Colab sessions. Total training time was approximately 45 minutes for both phases combined. The pytorch-grad-cam library (version 1.x) was used for Grad-CAM implementation.

6. Explainability — Grad-CAM

6.1 Why Explainability Matters in Healthcare

In high-stakes domains such as clinical medicine, a model's prediction is not sufficient on its own. Clinicians must be able to assess whether the model's reasoning is consistent with established medical knowledge before incorporating it into decision-making. A model that achieves 90% accuracy by attending to irrelevant image features such as patient markers, equipment artifacts, or demographic cues would be unsafe to deploy even if its aggregate performance metrics appear strong. Explainability methods provide a mechanism to audit model behavior on individual cases, detect failure modes, and build appropriate trust.

6.2 Grad-CAM Algorithm

Grad-CAM [2] generates a class-discriminative localization map using the gradients of the target class score y^c with respect to the feature map activations A^k of a convolutional layer. The algorithm proceeds as follows:

7. Forward pass the input image through the network to obtain the class score y^c
8. Backpropagate gradients of y^c with respect to feature maps A^k in the target layer (ResNet-50 layer4)
9. Compute importance weights: $\alpha_k = \text{global average pool of } (d y^c / d A^k)$
10. Compute weighted combination: $L_{\text{Grad-CAM}} = \text{ReLU}(\sum_k \alpha_k A^k)$
11. Upsample $L_{\text{Grad-CAM}}$ to input resolution (224 x 224) using bilinear interpolation
12. Apply jet colormap and alpha-blend with original image for visualization

The ReLU operation retains only regions with positive influence on the target class. The target layer is the output of ResNet-50's layer4, the final convolutional block before global average pooling, which provides the highest-level spatial feature representation at 7 x 7 spatial resolution.

6.3 Heatmap Interpretation

In the generated overlays, warm colors (red, orange, yellow) indicate regions that contributed most positively to the predicted class. Cool colors (blue, purple) indicate regions with low or negative influence. Clinically, for a pneumonia prediction, the heatmap should activate over the lung parenchyma, particularly the lower lobes where consolidation most commonly presents, rather than over the cardiac silhouette, diaphragm, or image borders.

6.4 Grad-CAM Visualizations

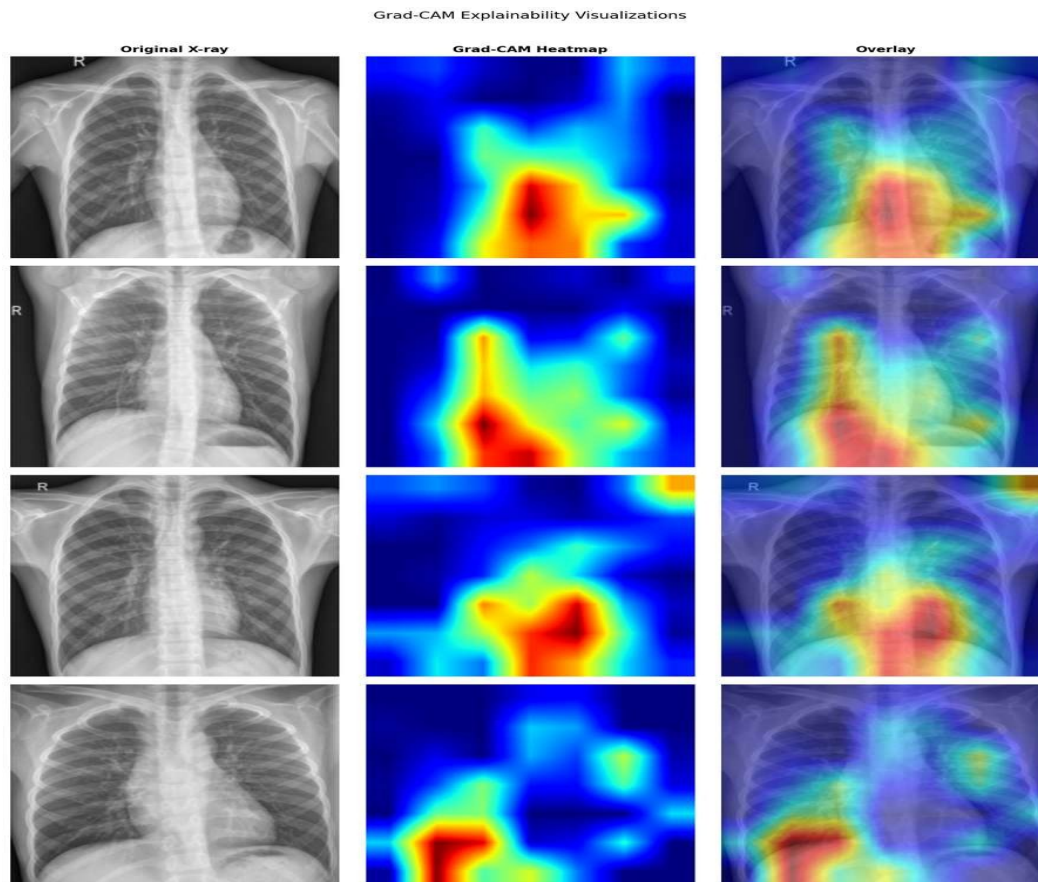


Figure 2: Grad-CAM explainability visualizations. Each row shows: original X-ray (left), raw Grad-CAM heatmap (center), and overlay on original image (right). Red regions indicate areas of highest influence on the model prediction. As shown in Figure 2, the model consistently activates over the lung fields rather than image artifacts or the cardiac border.

The heatmaps in Figure 2 demonstrate that the model consistently attends to the lung fields, with peak activations localized to regions of radiographic opacity consistent with pneumonia consolidation. This provides visual evidence that the model has learned clinically meaningful features rather than spurious correlates in the training data.

7. Results and Evaluation

7.1 Training Curves

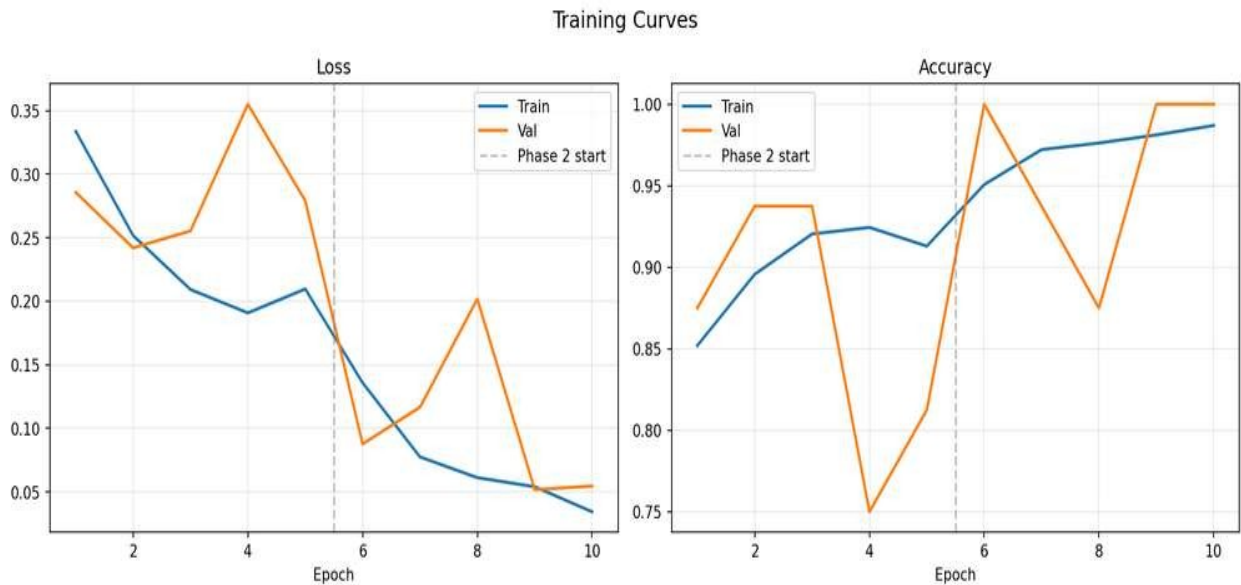


Figure 3: Training and validation loss (left) and accuracy (right) over 10 epochs. The dashed vertical line marks the transition from Phase 1 (head training) to Phase 2 (full fine-tuning). As shown in Figure 3, training loss decreases consistently across both phases, reaching approximately 0.04 by epoch 10.

The validation accuracy curve is noisy due to the small validation set size (16 images), which causes significant fluctuation. Despite this, the overall training trend is clear: loss converges well and the model does not overfit, as evidenced by the final test set performance reported in Section 7.2.

7.2 Test Set Performance

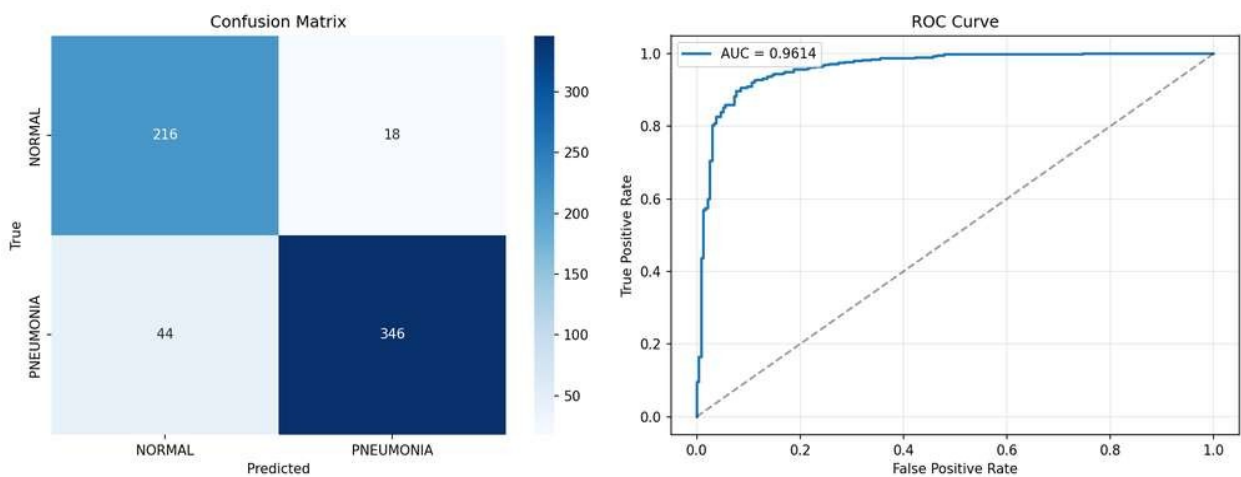


Figure 4: Confusion matrix (left) and ROC curve with AUC = 0.9614 (right) on the 624-image test set. As shown in Figure 4, the model achieves strong discriminative ability across all classification thresholds.

The confusion matrix in Figure 4 shows:

- 216 true negatives — Normal correctly classified as Normal
- 346 true positives — Pneumonia correctly classified as Pneumonia
- 18 false positives — Normal misclassified as Pneumonia
- 44 false negatives — Pneumonia misclassified as Normal

In a clinical context, false negatives carry greater risk than false positives, since a missed pneumonia case may result in delayed treatment. The model produces 44 false negatives versus 18 false positives, indicating it is appropriately conservative. The ROC-AUC of 0.9614 indicates excellent discriminative ability across all classification thresholds.

7.3 Per-Class Metrics

Class	Precision	Recall	F1-Score	Support
NORMAL	0.83	0.92	0.87	234
PNEUMONIA	0.95	0.89	0.92	390
Weighted Avg	0.91	0.90	0.90	624

Table 3: Per-class classification report on the 624-image test set

As shown in Table 3, the model performs comparably across both classes. Pneumonia precision of 0.95 indicates that when the model predicts pneumonia, it is correct 95% of the time. The weighted F1-score of 0.90 reflects strong overall performance across both classes despite the underlying class imbalance in the test set.

8. Challenges Faced

8.1 Class Imbalance

The 2.9:1 imbalance between pneumonia and normal cases in the training set posed a risk of the model developing a bias toward predicting pneumonia. Initial experiments without imbalance correction showed inflated overall accuracy alongside poor normal class recall. `WeightedRandomSampler` resolved this, but required careful configuration: the sampler weights must be computed from training set class frequencies and applied only to the training `DataLoader`, not validation or test loaders.

8.2 Validation Set Size

The Kaggle dataset's validation split contains only 16 images (8 per class), which is too small for stable validation metric estimation. This caused significant fluctuation in the validation accuracy curve during training, as visible in Figure 3, and made early stopping based on validation accuracy noisy. In practice, the training loss was used as the primary convergence indicator, with validation accuracy serving as a secondary signal.

8.3 Training Instability in Phase 2

Unfreezing all backbone layers in Phase 2 initially caused validation performance to drop before recovering. This is a known phenomenon in progressive fine-tuning: backbone parameters are being adapted from ImageNet representations to the target domain, which temporarily disrupts learned features. Differential learning rates ($1e-4$ for backbone, $1e-3$ for head) and the `CosineAnnealingLR` scheduler substantially reduced this instability compared to a fixed uniform learning rate.

8.4 Compute Limitations

Google Colab's free-tier GPU sessions are time-limited and can be interrupted without warning. Checkpointing to Google Drive after every epoch improvement was essential to prevent loss of training progress. Session restarts required re-running data loading and library installation cells, adding overhead. For longer training runs, a paid Colab tier or institutional HPC cluster would be preferable.

8.5 False Negatives

The model produced 44 false negatives, representing pneumonia cases classified as normal. Grad-CAM analysis of these cases, discussed in Section 9, revealed that many involve subtle, patchy, or viral pneumonia patterns that lack the dense consolidation characteristic of bacterial pneumonia. The training set contains predominantly bacterial pneumonia images, which limits generalization to atypical presentations.

9. Error Analysis with Grad-CAM

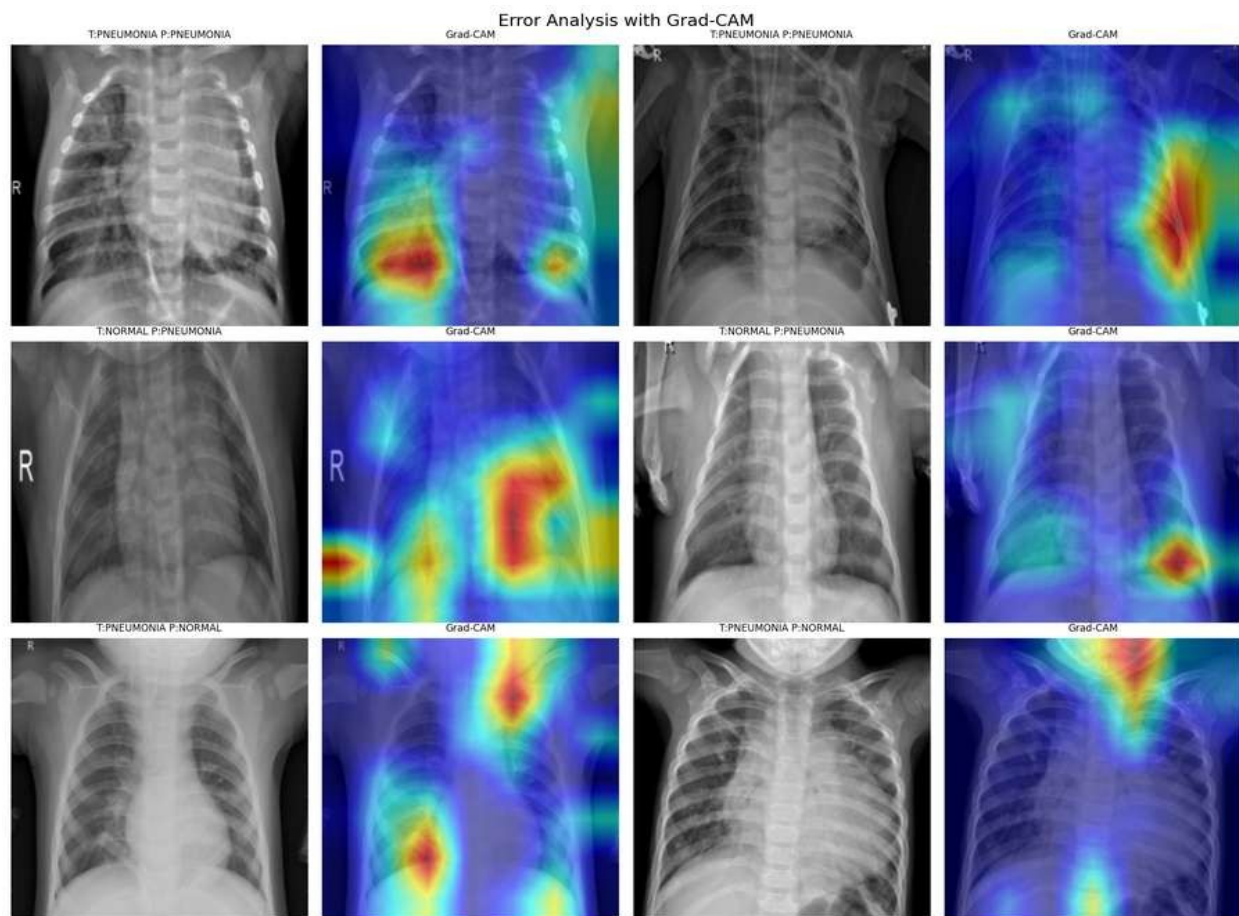


Figure 5: Grad-CAM error analysis. Row 1: True positives — Pneumonia correctly detected. Row 2: False positives — Normal misclassified as Pneumonia. Row 3: False negatives — Pneumonia cases missed. Each pair shows the original X-ray followed by the Grad-CAM overlay.

9.1 True Positives

As shown in Figure 5 (Row 1), in correctly detected pneumonia cases, Grad-CAM heatmaps show strong, localized activation over opacified lung tissue in the lower lobes, which is the anatomical region most commonly affected by lobar and bronchopneumonia. The spatial specificity of these activations provides evidence that the model has learned clinically meaningful features consistent with radiological presentation of bacterial pneumonia.

9.2 False Positives

As shown in Figure 5 (Row 2), false positive cases show activation concentrated near the hilar regions and along pulmonary vascular markings. The model appears to be confusing prominent pulmonary

vasculature or perihilar haziness with early-stage infiltrates. This is a recognized challenge even for human readers, as peribronchial thickening and increased vascular markings can mimic early pneumonia on plain film radiographs.

9.3 False Negatives

As shown in Figure 5 (Row 3), missed pneumonia cases show diffuse, low-intensity, and peripheral activation rather than centrally localized activation over areas of pathology. These cases appear to involve subtle interstitial patterns consistent with viral or atypical pneumonia, which may not produce the dense focal opacity that the model has learned to associate with pneumonia during training. This failure mode suggests the model would benefit from additional training data representing atypical pneumonia presentations.

10. Limitations

The following limitations apply to the current implementation and should be considered when interpreting results:

- **Binary classification only:** The system is trained and evaluated for binary Normal vs. Pneumonia classification. It does not generalize to the multi-label multi-pathology setting (e.g., pleural effusion, cardiomegaly, atelectasis) without retraining.
- **Pediatric-only dataset:** The Kaggle dataset originates exclusively from pediatric patients aged one to five years at a single institution. Performance on adult populations or images from different imaging systems and protocols has not been evaluated.
- **Small validation set:** The 16-image validation split is insufficient for reliable hyperparameter selection or early stopping. A larger, more representative validation set would improve training stability and model selection.
- **No clinical deployment validation:** The system has not been tested in a clinical workflow, with real-time image acquisition, DICOM formatting, or integration with existing radiology information systems.
- **Potential dataset bias:** Training predominantly on bacterial pneumonia images may limit sensitivity to viral and atypical presentations, as evidenced by the false negative analysis in Section 9.3.
- **Grad-CAM resolution:** Grad-CAM operates at the 7 x 7 feature map resolution of layer4, which limits spatial granularity of the resulting heatmap. Finer localization methods such as Grad-CAM++ or Score-CAM may provide more precise region attribution.

11. Future Work

11.1 Multi-Class Detection

The most direct extension is multi-class pathology classification using the NIH ChestX-ray14 dataset [4], which provides 14 disease categories across over 100,000 images. This would require a multi-label classification head and per-class Grad-CAM analysis to evaluate localization quality for each pathology.

11.2 Improved Explainability Methods

Grad-CAM is a class-discriminative but coarse localization method. More granular alternatives include Grad-CAM++ [Chattopadhyay et al., 2018], which provides better localization for multiple object instances, and Score-CAM [Wang et al., 2020], which eliminates gradient dependence entirely. SHAP-based pixel-level attribution could also be explored for finer-grained explanations.

11.3 Clinical Validation

A prospective reader study comparing model predictions and Grad-CAM heatmaps against radiologist diagnoses would provide evidence of clinical utility. Such a study would require IRB approval, patient recruitment, and collaboration with a clinical institution, and would be a necessary step before any deployment in a care setting.

11.4 Addressing Atypical Pneumonia

The high false negative rate for subtle viral pneumonia presentations suggests the model would benefit from additional training examples of atypical patterns, including COVID-19 chest X-ray datasets and pediatric viral pneumonia cases from diverse clinical settings.

11.5 Mobile and Edge Deployment

The TorchScript-traced model produced in this project (model_scripted.pt) is a foundation for mobile deployment. Post-training quantization or quantization-aware training could reduce model size by up to 4x, enabling deployment on resource-constrained devices in low-infrastructure settings.

12. Conclusion

This project demonstrates a complete, end-to-end explainable deep learning pipeline for automated pneumonia detection from chest X-ray images. Using a two-phase transfer learning strategy on ResNet-50, the system achieves 90.1% test accuracy and a ROC-AUC of 0.9614 on the Kaggle Chest X-Ray dataset. Grad-CAM visualizations confirm that the model attends to clinically relevant lung regions rather than image artifacts, providing a critical layer of transparency for potential clinical use. Systematic error analysis using Grad-CAM on true positives, false positives, and false negatives provides interpretable insight into model failure modes and suggests concrete directions for improvement.

From a technical standpoint, this project delivers a reproducible training pipeline with documented hyperparameters, a structured evaluation framework including confusion matrix and ROC analysis, and an integrated explainability module. These components are deployed as a publicly accessible Gradio web application, demonstrating that explainability can be embedded as a first-class design requirement rather than an afterthought in medical AI systems.

The limitations of this work, including binary classification scope, a pediatric-only dataset, and the absence of clinical validation, are clearly identified and directly motivate the future work outlined above. As healthcare AI continues to mature, the integration of transparency, interpretability, and rigorous evaluation will be essential to building systems that clinicians can trust and patients can benefit from.

Project Resources

Project Repository:

<https://github.com/ameerascsu/Explainable-Deep-Learning-System-for-Chest-X-Ray-Classification>

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Appendix A — Additional Figures and Tables

A.1 Confusion Matrix and ROC Curve

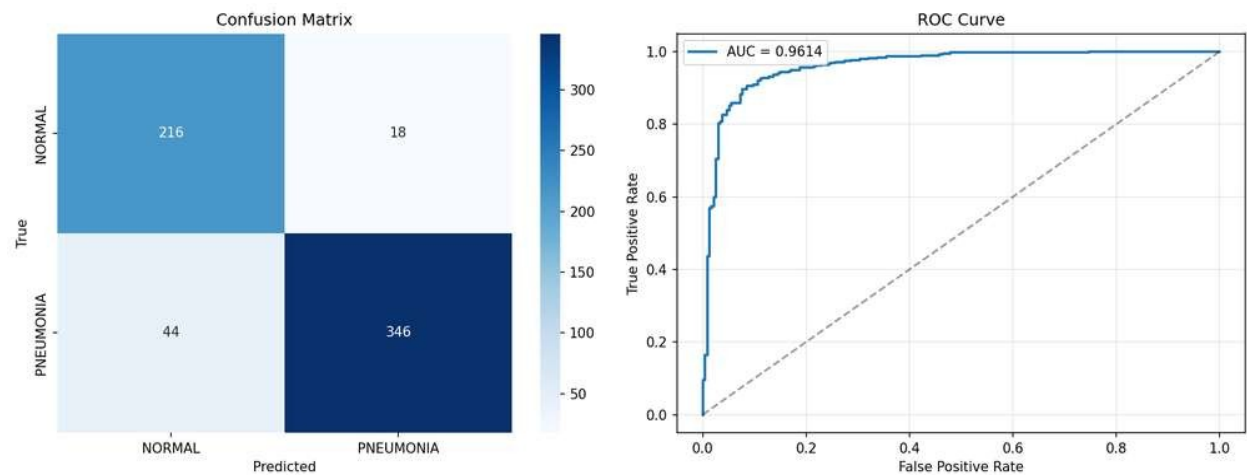


Figure A.1: Full evaluation results on test set (624 images). Confusion matrix shows 216 TN, 18 FP, 44 FN, 346 TP. ROC-AUC = 0.9614.

A.2 Training Curves

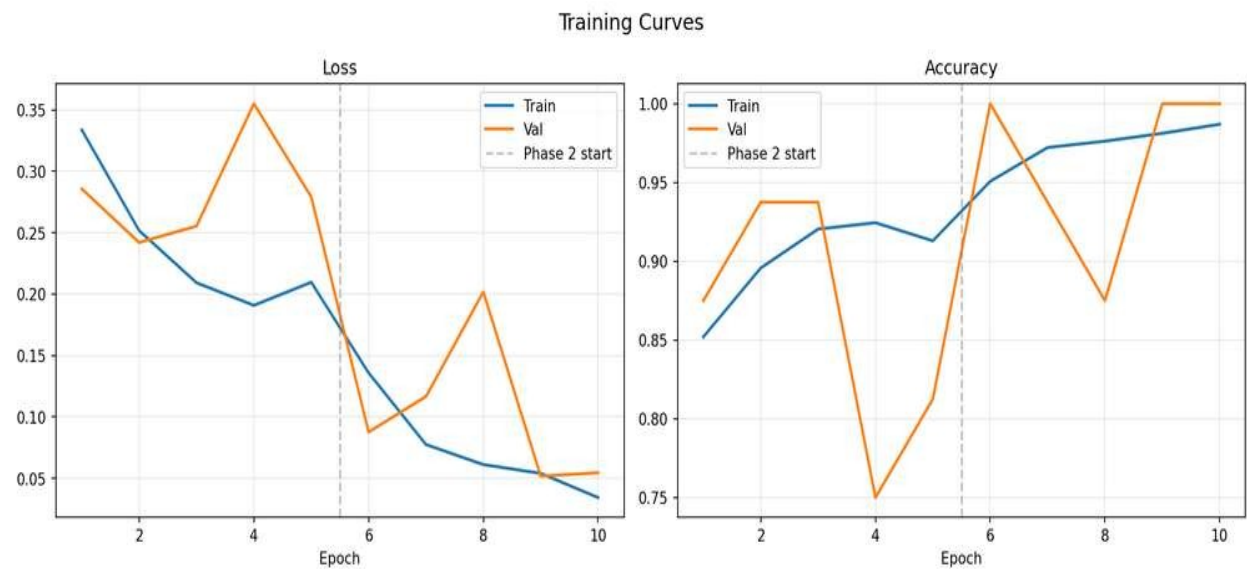


Figure A.2: Training and validation loss and accuracy across 10 epochs. Dashed line marks Phase 2 start.

A.3 Grad-CAM Explainability Outputs

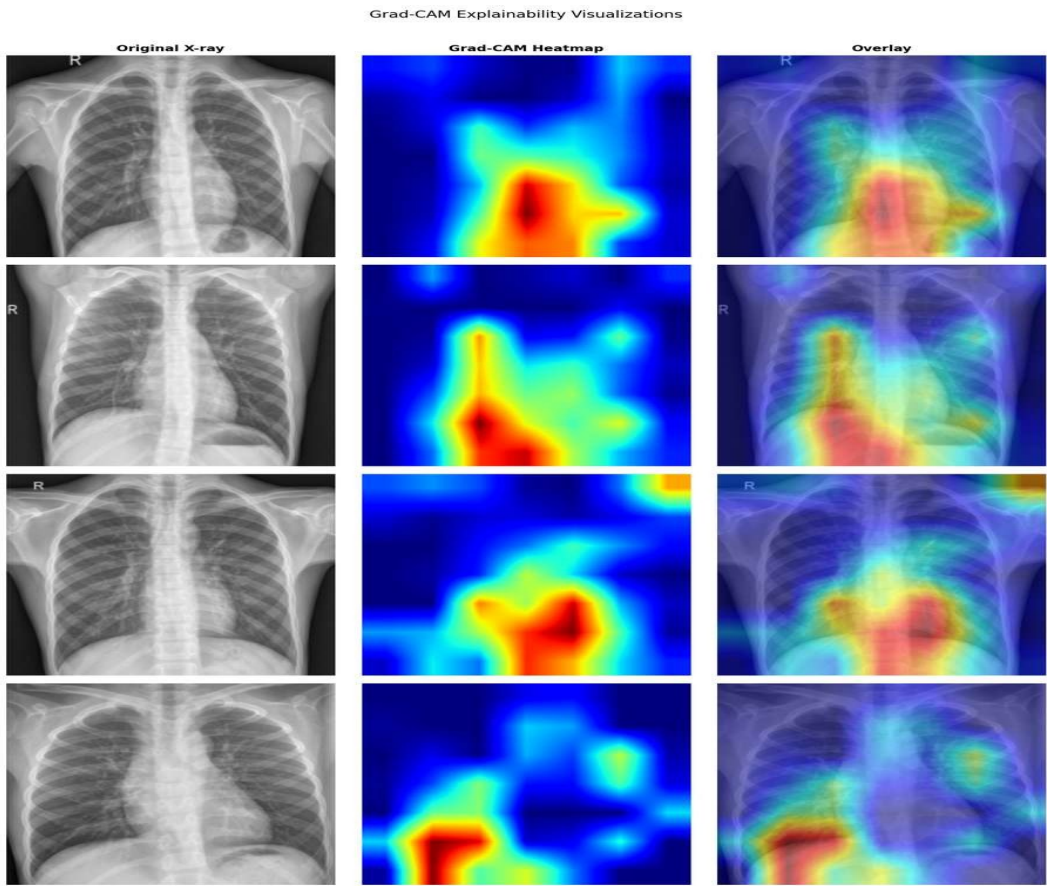


Figure A.3: Sample Grad-CAM explainability outputs showing original X-ray, heatmap, and overlay for four test cases.

A.4 Grad-CAM Error Analysis

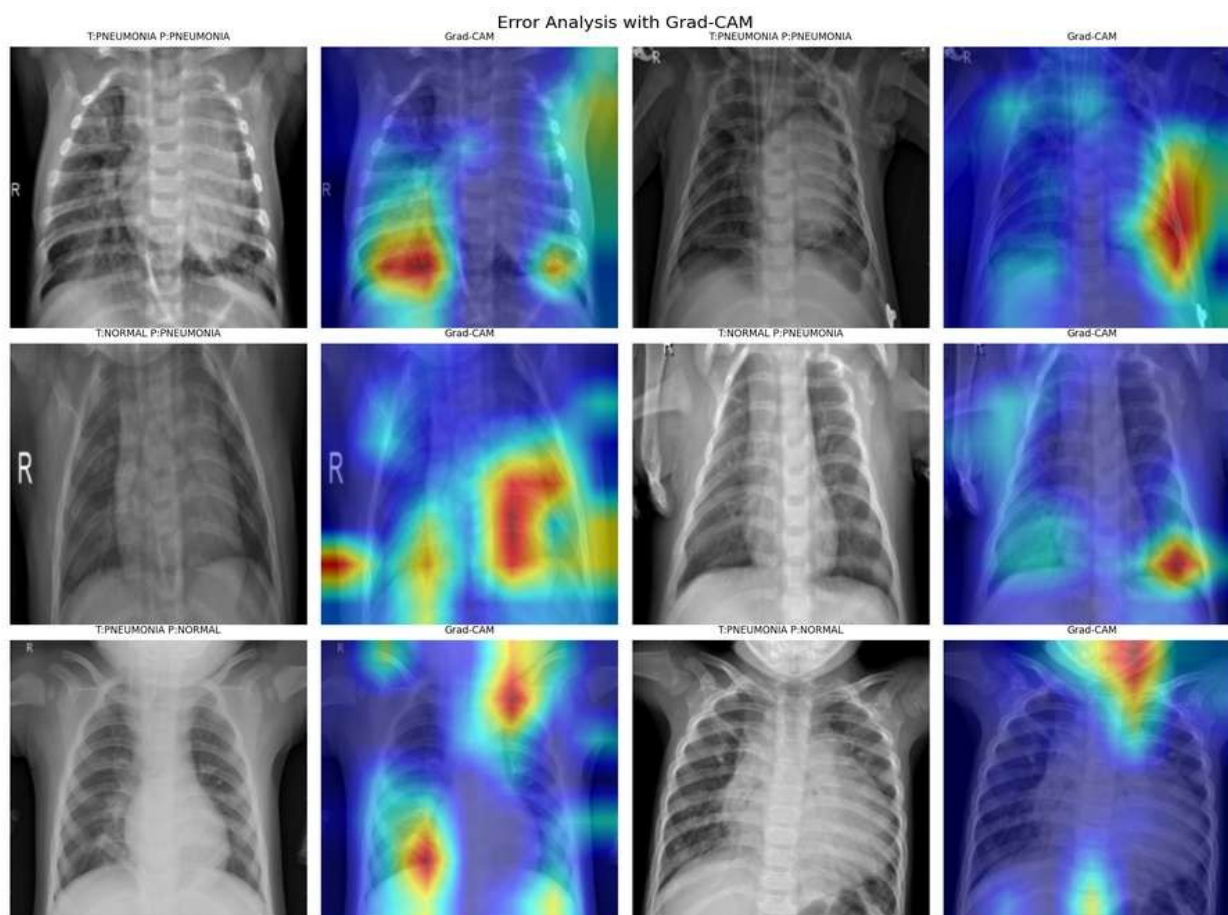


Figure A.4: Grad-CAM error analysis across true positives (Row 1), false positives (Row 2), and false negatives (Row 3).

A.5 Project File Structure

```
pulmonary_disease_detection.ipynb | +-- Cell 01-02: Environment setup and Kaggle
credentials +-- Cell 03: Dataset download (chest-xray-pneumonia) +-- Cell 04:
Google Drive mount for checkpoint persistence +-- Cell 05: Data preprocessing
pipeline and DataLoaders +-- Cell 06: Sample visualization (8 training images)
+-- Cell 07: ResNet-50 model architecture definition +-- Cell 08: Training
and evaluation utility functions +-- Cell 09: Phase 1 training (head only, 5
epochs) +-- Cell 10: Phase 2 fine-tuning (all layers, up to 10 epochs) +--
Cell 11: Training curve plots (loss and accuracy) +-- Cell 12: Test set
evaluation (metrics, confusion matrix, ROC) +-- Cell 13: Grad-CAM module
initialization +-- Cell 14: Grad-CAM visualization on test samples +-- Cell
15: Error analysis (TP, FP, FN with Grad-CAM) +-- Cell 16: Model export
(best_model.pth, model_scripted.pt) +-- Cell 17: Gradio deployment (Hugging
Face Spaces) Google Drive output/ +-- best_model.pth (best validation
checkpoint) +-- model_scripted.pt (TorchScript for deployment) +--
model_info.json (architecture and metric metadata) +-- training_curves.png
+-- evaluation.png +-- gradcam_results.png +-- error_analysis.png
```

A.6 Dependencies

Library	Version	Purpose
PyTorch	2.x	Deep learning framework
torchvision	0.x	Pretrained models & transforms
grad-cam	1.x	Grad-CAM implementation
scikit-learn	1.x	Evaluation metrics
matplotlib	3.x	Visualization
seaborn	0.x	Statistical plots
Gradio	4.x	Web application interface
torchmetrics	1.x	Metric computation

Table A.1: Python library dependencies